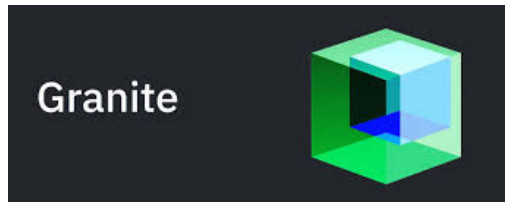


IBM's AI models set new standards for enterprise AI performance

IBM has announced the launch of Granite 3.0, its latest suite of AI models. This release introduces powerful language models and specialised tools, all available under the open Apache 2.0 license, setting a new benchmark for enterprise AI performance, safety, and flexibility.

Granite 3.0 8B and 2B models excel in core AI tasks like retrieval augmented generation (RAG), classification, and summarisation. Mixture-of-Experts (MoE) models offer efficient inference with low latency, optimised for CPU-based deployments and edge computing. The Granite Guardian 3.0 models deliver the company's most advanced



safety features to detect risks such as bias, toxicity, and hallucinations. These models supposedly outperform Meta's Llama Guard models across several safety benchmarks.

The updated time-series models achieve state-of-the-art results in zero/few-shot forecasting, surpassing models 10 times larger from Google and Alibaba. The models will power the company's Consulting Advantage, boosting productivity for its 160,000 consultants. They are integrated into platforms like Hugging Face, NVIDIA NIM microservices, and Google Vertex AI, supporting edge deployments through tools like Ollama and Replicate.

GNOME Mutter adopts high-priority KMS thread to prevent crashes

The GNOME Mutter compositor has updated its kernel mode-setting (KMS) thread scheduling, shifting from real-time to high-priority status. This change addresses stability issues that caused crashes and unexpected process terminations in GNOME Shell and Mutter.

This adjustment resolves numerous bugs reported over the past year, including issues like OpenGL context loss, crashes, freezes, and GNOME Shell terminations during random actions, particularly in QEMU virtual machines. With the newly committed code, many crash reports and related bugs are now considered resolved.

This improvement highlights ongoing efforts to enhance the reliability and performance of the GNOME environment, benefiting both developers and end users.

Jonas Adahl of Red Hat elaborated on the changeover to using a high priority KMS thread: "In contrast to real-time scheduling, this doesn't risk us getting SIGKILL:ed when the kernel is doing busy looping in `drmModeAtomicCommit()` for some reason, but will according to testing, right now, give us more or less the same benefit when it comes to dispatch lateness and commit lateness."

Linux kernel 6.12 released: Real-time capabilities and Raspberry Pi 5 support added

The Linux kernel 6.12 is now available, bringing numerous enhancements and marking its status as a Long-Term Support (LTS) release, with updates assured through 2026.

This latest version introduces several significant features. It comes with enhanced support for real-time applications, making the kernel better suited for systems requiring precise timing and responsiveness. It also has full compatibility with the latest Raspberry Pi 5 hardware, expanding its usability in IoT, education, and hobbyist projects. Improved drivers and networking capabilities ensure better

AlmaLinux OS 9.5 released as a free alternative to RHEL 9.5

The AlmaLinux OS Foundation has announced the release of AlmaLinux OS 9.5 (Teal Serval), the latest stable version of its free, open source alternative to Red Hat Enterprise Linux (RHEL).

Built from the same sources as RHEL 9.5, AlmaLinux OS 9.5 ensures full compatibility with RHEL while introducing several improvements, including better system performance monitoring, visualisation, and data collection.

AlmaLinux OS 9.5 includes updated module streams for improved support for web apps, enhanced cryptography with stronger security updates and updates to crypto-policies for better algorithm control in Java, and an updated SELinux policy to enable QEMU Guest Agent to execute confined commands.

Performance, reliability, and scalability for developers have been boosted with new versions of tools such as GCC 14, LLVM 18.1.8, Rust 1.79.0, and Go 1.22. Other updated components include .NET 9.0, Apache 2.4.62, Node.js 22, and OpenSSL 3.2.2, among others.

Like RHEL 9.5, AlmaLinux OS 9.5 is powered by Linux kernel 5.14.0-427.13.1.el9_4. This release is available for download as 64-bit (x86_64), AArch64 (ARM64), ppc64le, and s390x installation ISOs from the official website.

